

Serious Materials Rocks

Gypsum. It's not just for Twinkies. The mineral, a semi hydrous form of calcium sulfate, with many uses from Plaster of Paris to Chinese Medicine and even Twinkies, is mined all over the world. It's most popular use is in the most prevalent of the building materials, sheetrock, or drywall. Invented in 1917, when energy was cheap, sheetrock is like other energy-intensive concoctions of its time, a miracle invention that revolutionized home-building, making it much easier than prevailing practices. But its manufacture owes a full 25% to 50% of its costs to energy. It joins its sister materials glass, steel, and cement in being a major part of the problem of greenhouse gases.

To make sheetrock, gypsum is heated in large natural gas kilns to about 300°F (coughing lots of CO₂ into the sky), and then dried (again, using natural gas). Additives are mixed in such as fiberglass and mold repellants, and then paper backing is placed on one side of the dry plaster, as well as the manufacturer's stamp. Considering its prevalence in the building industries, and the building industries' prevalence in worldwide energy consumption, the evolution of drywall could play a major role in the reduction of greenhouse gases. What if the basic materials and processes were entirely rethought?

"Drywall production consumes almost 1% of all U.S. industrial energy consumption—creating 25 billion pounds of greenhouse gases in the process," says Serious Materials, in California's Silicon Valley, makers of several green building materials such as soundproofing wallboard, noise-reduction coatings, super-insulating windows, and solar panels. It is now producing drywall in low-impact ways, by using only a little electricity in the manufacture, and no fossil fuels. Most remarkably it doesn't use gypsum!



EcoRock: Most remarkably, it doesn't use gypsum.

SERIOUS MATERIALS

Based in Sunnyvale, Serious Materials takes its edge from the innovative spirit of the dot com generation. Its founders knew that if they could reinvent the material without reinventing the peripheral equipment, uses and processes, they could change the paradigm in a significant way. Its first EcoRock factory is in Sunnyvale California, and takes up its space in an empty warehouse in a process that doesn't require the use of any natural gas, and only a little electricity, thus there are no greenhouse gases produced in the making of the wallboard, which acts just like conventional wallboard. Its manufacture takes 90% less energy to produce than conventional drywall. It hangs the same; it can be recycled just like sheetrock, but even better, it can biodegrade, and is lighter, reducing the environmental costs of shipping.

According to V.P. of Marketing, Steve Weiss, "Eco Rock is made from commonly available, eco-friendly materials that are fully recyclable at end of life, and of highly recycled content to begin with." And what is that material? You had to ask. EcoRock's basic materials (a secret, for now), begin in powder form, which, when put together, heat up, giving off the liquid part, and then set in just minutes. Weiss says, "Conceptually that's what's going on and that's wildly different. When opening up our plants there are ceremonies in which we cap off the gas mains that come into these old factories, because we don't

need them. Which means factories can be located lots of places that are far less expensive to build in, in terms of permitting. This is especially good because some places have already outlawed adding plants which generate more CO₂."

Serious Materials's EcoRock has more than paper backing. They have the confidence of several investors like New Enterprise Associates, Foundation Capital, and Rustic Canyon Partners, with commitments of over \$50 million to get the new board off the ground (and on the walls). "Drywall is a straight-forward, familiar, and useful thing," says Weiss. "Over time there may be eco-roofing and eco-siding and eco-decking. Things that adopt zero emissions processes. We might consider the old stuff barbaric."

Stand by for EcoRock's release this fall, 2008. Look for Serious Materials other products, such as ThermaProof windows and QuietRock, soundproof drywall. **He**

—Leslie Jackson

Leslie Jackson is an associate editor at Home Energy.

For more information:

For more about Serious Materials, go to www.seriousmaterials.com.

To see Serious Materials' Quiet Solution line, go to www.quietsolution.com.